

Powermax45 SYNC

Recommended cut capacity of 16 mm (5/8")

Severance capacity of 29 mm (1 1/8")

Professional grade



Part of the Powermax® family, the Powermax45 SYNC features advanced torch communication for automated setup and patented single-piece cartridges for hand cutting, mechanized or robotic cutting, marking, or gouging.

Key business benefits

Simplify operations and boost productivity

Single-piece cartridge simplifies operations and dramatically reduces training time

Minimize operating costs

Up to 5x longer consumable life compared to competitors decreases operating costs

Reduce secondary operations

Improved cut quality over consumable life delivers better part consistency to reduce secondary operations

Powermax45 SYNC cut and gouge capacities

Cut capacity	Thickness	Cut speed
Recommended	16 mm (5/8")	500 mm/min (20 ipm)
Severance (hand cutting)	29 mm (1-1/8")	125 mm/min (5 ipm)
Pierce*	12 mm (1/2")	
*Pierce rating for handheld use or with automatic torch height control		
Gouge capacity	Metal removal rate	Groove profile
Typical gouge	3.4 kg (7.5 lbs) per hour	3.2 mm (.12") D x 6,8 mm (.26") W

Torch output**	6.9 kW
	45 A x 155 V ÷ 1000 W = 6.9 kW

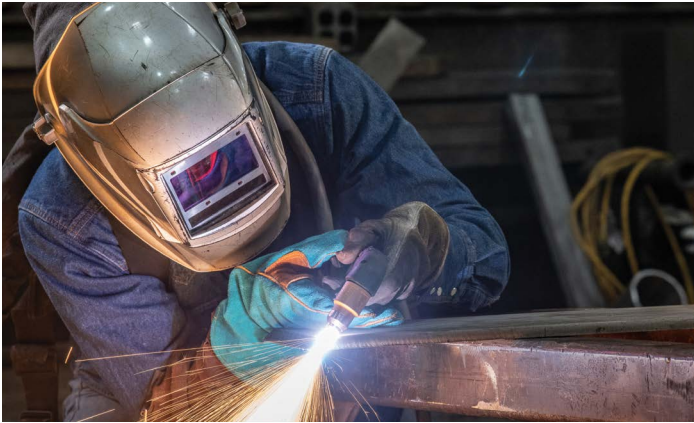
**Higher output voltage at the torch plus efficiently designed consumables result in a faster, cleaner cut.

Powermax45 SYNC. Easy to use. Versatile. Reliable.

For metal workers who want the most versatile system available to gouge, cut, and mark, Powermax45 SYNC delivers the best return on investment with a patented single-piece cartridge, up to 5x longer consumable life, and greater cutting power than higher-amperage competitors.

Our proprietary single-piece, color-coded cartridges help operators eliminate mix-ups, simplify inventory management, and decrease operating costs.

**THE BEST
INVESTMENT
IS IN YOUR
POWER**



Highlighted product features:

Easy to use

- A patented single-piece cartridge eliminates operator error and simplifies operations
- Advanced torch communication for automated setup improves cut quality

- Easy to set up for mechanized usage with optional CNC interface and FastConnect™ torch connection

Versatile

- Gain greater cutting power than higher-amperage competitors
- Expand application capabilities with a wide variety of specialty cartridges and torches
- Cut and mark using the same FineCut® cartridge

Industry-leading reliability

- Powermax plasma cutters are known to last decades in the field, minimizing maintenance and downtime
- SmartSYNC® torches are engineered for rugged industrial environments
- Each Powermax component is tested to perform reliably in demanding conditions

Our customers' preferred cutting tool for industries including:

- Job shops/fabricators
- Property/plant maintenance
- HVAC/mechanical contracting
- Educational/trade schools
- Agricultural/equipment maintenance

Specifications

CE – for use in Europe, Australia & New Zealand – RCM, Serbia, UK – UKCA, Ukraine, and other countries where accepted.

CCC – for use in China

CSA – for use in the Americas and Asia, except China

CE	CCC	CSA	Specifications
Input voltages (± 10%)			
		↗	200–240 V, 1-PH, 50–60 Hz
		↘	480 V, 3-PH, 50–60 Hz1
↗	↘		230 V, 1-PH, 50–60 Hz
↗	↘		400 V, 3-PH, 50–60 Hz
Input current @ 6.9 kW			
		↗	200–240 V, 1-PH, 39/32 A
↗	↘		230 V, 1-PH, 33 A
↗	↘		400 V, 3-PH, 10 A
Output current			9–45 A
Rated output voltage			155 VDC
Duty cycle @ 40° C (104° F)			
		↗	50% @ 45 A, 200–240 V, 1-PH
		↘	60% @ 41 A, 200–240 V, 1-PH
		↗	100% @ 32 A, 200–240 V, 1-PH
		↘	50% @ 45 A, 480 V, 3-PH
		↘	60% @ 41 A, 480 V, 3-PH
		↗	100% @ 32 A, 480 V, 3-PH
↗	↘		50% @ 45 A, 230 V, 1-PH
↗	↘		60% @ 41 A, 230 V, 1-PH
↗	↘		100% @ 32 A, 230 V, 1-PH
↗	↘		50% @ 45 A, 400 V, 3-PH
↗	↘		60% @ 41 A, 400 V, 3-PH
↗	↘		100% @ 32 A, 400 V, 3-PH
Open circuit voltage (OCV)			
↗	↘	↗	275 VDC 200–240 V
		↘	275 VDC 480 V
↗	↘		265 VDC 400 V
Dimensions with handles			442 mm (17.4") D; 173 mm (6.8") W; 357 mm (14.1") H
Weight with 6.1 m (20') torch			
		↗	14 kg (31 lbs.)
↗	↘		15 kg (33 lbs.)
Gas supply			Cutting: air (clean, dry and oil-free), nitrogen, F5 Gouging: air (clean, dry and oil-free), nitrogen, F5 Marking: air (clean dry and oil-free)
Recommended gas inlet flow rate/pressure			188 l/min @ 5.9 bar (400 scfh 6.7 scfm @ 90 psi)
Input power cable length			3 m (10')
Power supply type			Inverter – IGBT
Engine drive requirement			12.5 kVA (10 kW) for full 45 A output
Electrical efficiency			90%
Recyclability			100%
Warranty			Power supply: 3-year, Torch: 1-year

Ordering information

Below are the standard system configurations, which include a power supply, torch, and work cable.

CE – for use in Europe, Australia & New Zealand - RCM, Serbia, UK – UKCA, Ukraine, and other countries where accepted. CCC – for use in China CSA – for use in the Americas and Asia, except China				75° hand torch		75° and 15° hand torch	180° full-length machine torch		180° full-length machine torch and 75° hand torch
CE	CCC	CSA	Torch lead length	6.1 m (20')	15.2 m (50')	6.1 m (20') and 7.6 m (25')	7.6 m (25')	15.2 m (50')	7.6 m (25') and 6.1 m (20')
✓ = With remote pendant							✓		✓

Powermax45 SYNC® power supplies				Handheld			Mechanized			
		↳	230V CSA standard power supply	088560	088561	088564				
		↳	230V CSA power supply with CPC port and voltage divider	088562	088563		088580	088582	088581	088583
		↳	230V CSA power supply with CPC port, voltage divider board, and serial port				088584			
↳	↳		400V CE/CCC standard power supply	088565	088566					
↳	↳		400V CE/CCC power supply with CPC port and voltage divider	088567			088585		088586	
↳	↳		230V CE/CCC standard power supply	088568						
↳	↳		230V CE/CCC power supply with CPC port and voltage divider	088569			088587			

Custom configurations

Select power supply, torch, work cable, cartridge, and other components.

Power supply options

CE – for use in Europe, Australia & New Zealand - RCM, Serbia, UK – UKCA, Ukraine, and other countries where accepted. CCC –for use in China CSA – for use in the Americas and Asia, except China				Standard	With CPC port and voltage divider	With CPC port, voltage divider and serial port
CE	CCC	CSA				

		↳	Powermax45 SYNC 230V CSA	088570	088571	088572
		↳	Powermax45 SYNC 480V CSA	088573		
↳	↳		Powermax45 SYNC 400V CE/CCC	088574	088575	088576
↳	↳		Powermax45 SYNC 230V CE/CCC	088577	088578	088579

Torch options

	Hand torches		Long torches				Machine torches	Robotic/Mini		
	75°	15°	0.6 m (2'), 45°	0.6 m (2'), 90°	1.2 m (4'), 45°	1.2 m (4'), 90°	180°	180°	90°	45°
4.5 m (15')								059733		
7.6 m (25')	059726	059723			528114		059719	059734	059731	059729
10.7 m (35')							059720			
15.2 m (50')	059727	059724	528116	528117	528118	528119	059721	059735	059732	059730
22.8 m (75')	059728	059725					059722		059767	059766
30 m (100')	059770									

Cable options

	Work cables	Control cables				
Cable length	Hand clamp	Remote pendant	CNC, spade connector, divided voltage	CNC, spade connector, no divided voltage	RS-485 Serial comm, unterminated	RS-485 Serial comm, D-sub
7.6 m (25')	223595	128650	228350	023206	223236	223239
15.2 m (50')	223596	128651	228351	023279	223237	223240
22.8 m (75')	223127	128652				

Powermax cartridge options

	Drag cutting	Mechanized/standoff cutting	Max removal gouging	Max control gouging	Ohmic ring kit
45 A	428927	428925	428932	428933	428895

	FineCut®	
	Handheld	Mechanized
30-45 A	428928	428926

Accessories

Part Number	Description
528083	Hypertherm Cartridge Reader
128647	Eliminizer Air Filtration Kit, 1-micron filter and auto-drain moisture separator
024877	Leather Torch Lead Cover with hook and loop closure, Black, 7.6 m (25') section
027669	Plasma Circle Cutting Guide, SmartSYNC Deluxe Kit with magnetic base
127219	System Storage Dust Cover, Powermax45/45 XP/45 SYNC

Torches and cartridges



Learn more at www.hypertherm.com/powermax45sync

Unless otherwise noted in the collateral, all trademarks are property of Hypertherm, Inc. and may be registered in the United States and/or other countries.

Please visit www.hypertherm.com/patents for more details about Hypertherm Associates patent numbers and types.

© 4/2024 Hypertherm, Inc. Revision 1

898550

As 100% Associate owners, we are all focused on delivering a superior customer experience. www.hyperthermassociates.com/ownership

Environmental stewardship is one of Hypertherm Associates' core values. www.hyperthermassociates.com/environment

100% Associate-owned

